Explainable AI and SHAP: Understanding and Explaining Machine Learning Models

The Stakes of AI Decisions

AI systems now make high-stakes decisions.

Domain	Decision	Impact
Finance	Loan approval, credit scoring	Economic opportunity
Healthcare	Diagnosis, treatment recommendations	Patient outcomes
Criminal Justice	Risk assessment, sentencing	Liberty and fairness
Hiring	Resume screening, interview scoring	Employment opportunity

The question is not just:

"What did the model predict?"

Instead, we ask:

- "Why?"
- "Should we trust it?"

What is Explainable AI?

Explainable AI (XAI) refers to methods that make AI system outputs understandable to humans.

Goal	Question
Transparency	How does the model work? What patterns did it learn?
Interpretability	Why did it make *this* prediction? Which features mattered?
Trustworthiness	Is the reasoning sound? Does it align with domain knowledge?

XAI bridges the gap between model performance and human understanding.

The XAI Landscape

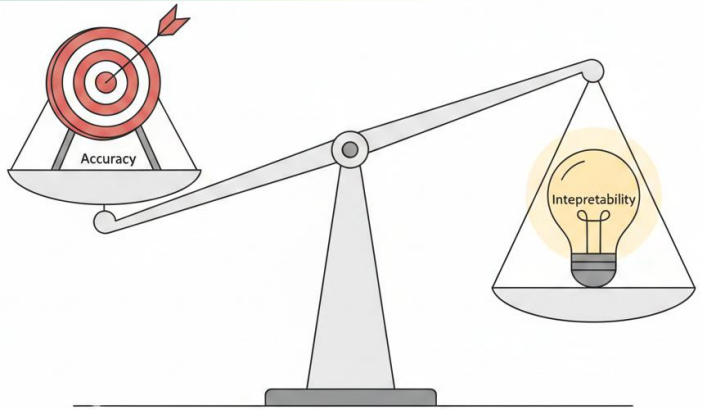
Different stakeholders ask different questions and require different types of explanations.

Stakeholder	Question	Needs
End User	"Why was I denied?"	Simple, actionable reasoning
Data Scientist	"Is this model learning spurious correlations?"	Debugging tools
Regulator	"Is this system fair?"	Auditable decision logic

Explanation methods differ in three main ways:

Dimension	Options	Why it Matters
Scope: What does it explain?	Global (overall behavior) vs. Local (single prediction)	Global: "Which features drive this model?" Local: "Why was this applicant denied?"
Model Access: Does it need access to model internals?	Model-agnostic (any model) vs. Model-specific	Agnostic works on any model; specific is faster but limited to certain architectures.
Timing: When is interpretability added?	Intrinsic (built-in) vs. Post-hoc (after training)	Intrinsic models (e.g., linear regression) are simple to understand directly but may be less accurate. Post-hoc methods let you use powerful models first, then explain them afterward.

The Problem with Simple Models



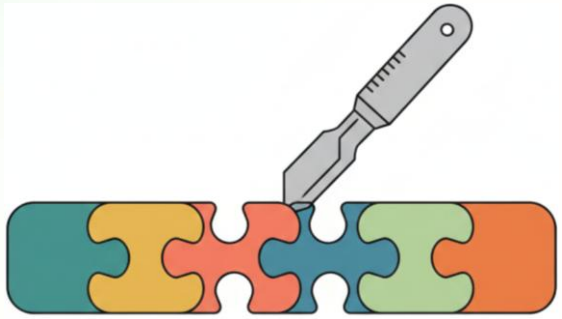
We could just use interpretable models...

Model	How to Interpret	Limitation
Linear regression	Coefficients are feature weights	Can't capture nonlinear relationships
Decision tree	Follow the path from root to leaf	Deep trees overfit; shallow trees underfit
Rule lists	Read the if-then rules	Struggles with continuous features

But we often need complex models (random forests, neural nets) for accuracy.

Can we get the accuracy of complex models and still explain their predictions?
(The post-hoc approach.)

What We Want from an Explanation



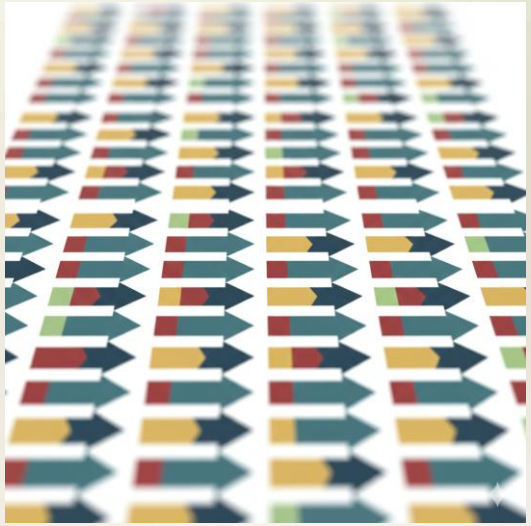
Features

For a single prediction, we want to know:

- How much did each feature contribute?
- Did it push the prediction up or down?
- Do the contributions add up to the final prediction? (Nothing left unexplained.)

Across many predictions, we want:

- Which features matter most overall?
- Consistent methodology: if a feature has the same effect in two different predictions, it should receive the same credit in both.



Is there a principled way to do this, not just a heuristic?

Revisiting Game Theory

The feature attribution problem looks like a fair division problem.

Fair Division

How a shared outcome is split among contributors, so each gets credit proportional to what they actually added.

A prediction is like a team's total earnings.
Features are like team members.
We want to divide credit fairly.

Shapley (1953) solved the fair division problem in economics; Lundberg & Lee (2017) applied it to machine learning.

SHAP = SHapley Additive exPlanations

A principled, theoretically grounded method for feature attribution.

Game Theory	→	SHAP
Players	→	Features
Payout	→	Prediction
Fair share	→	SHAP value

SHapley Additive exPlanations (SHAP)

An Analogy from Cooperative Game Theory

THE GAME

Three friends (Alice, Bob, Carol) collaborate and earn $100. How should they split it fairly?

SHAPLEY'S INSIGHT (1953):

Pay each player their average marginal contribution across all possible orderings of team formation.

Marginal is being used in the economics sense:

marginal ≡ "the incremental value of adding this player/feature to what is already there"

If Alice joins first and earns $40, then Bob joins (+$30), then Carol (+$30)…

but what if Bob joined first? Average over *all* arrival orders.

The SHAP Decomposition

Final prediction SHAP value for a single instance $\mathbf{x}$.

This formula defines a ***local scope*** by explaining one prediction.

$$f(\mathbf{x}) = \phi_0 + \phi_1 + \phi_2 + \cdots + \phi_n$$

The model's average output over all training examples: baseline

Feature i's SHAP value (ϕ_i) contribution

Assume the average applicant in the training data had 50% approval. What about this applicant pushed them to 78%?

$$f(\mathbf{x}) = 0.5 + 0.19 + 0.115 - 0.025 = 0.78$$

	Income	Credit Score	Debt Ratio	Model Prediction
	85K	720	35%	78% Approval
ϕ_i	0.19	0.115	-0.025	

How much did this feature push the prediction up or down from the baseline?

How do we compute ϕ_i?

Why Not Just Measure Each Feature Independently?

Idea #1

"Measure each feature's effect in isolation": hold all other features at neutral values (e.g., their averages), vary only one feature, and see how the prediction changes.

This fails because features are rarely independent.

A 720 (out of 800) credit score has a different effect depending on income: a high credit score may add little to the approval when an applicant may already have a high income which adds significantly.

Measuring in isolation assumes no interaction among features, producing a main effect that ignores how features combine in actual predictions.

Idea #2

Add features one at a time, tracking the changes.

Consider an ordering of features to add in sequence:
Credit → Income → Debt

Step	Add Feature	Prediction	Marginal
0	(none)	0.50	—
1	`Credit`	0.65	+0.15
2	`Income`	0.72	+0.07
3	`Debt`	0.78	+0.06

The problem is that the order matters. `Credit` arrives first and absorbs all shared effects. Income and Debt only get attribution for what remains after Credit Score already claimed the correlated impact.

If we used a different ordering we would get different attributions. Which ordering of features is "right"?

The Shapley Insight: Average Over All Orderings

What if we tried every possible ordering and averaged?

With 3 features how many orderings do we have?

$$P(3,3) = 3! = 6$$

n features?

$$n!$$

Ordering	Credit's marginal	Income's Marginal	Debt's marginal	Σ
C → I → D	+0.15	+0.07	+0.06	+0.28
C → D → I	+0.15	+0.18	-0.05	+0.28
I → C → D	+0.14	+0.08	+0.06	+0.28
I → D → C	+0.26	+0.08	-0.06	+0.28
D → C → I	+0.18	+0.18	-0.08	+0.28
D → I → C	+0.26	+0.10	-0.08	+0.28
Average: Shapley Value (ϕ_i)	+0.19	+0.115	-0.025	

IDEA: Average each column to get that feature's Shapley value thus no feature gets positional advantage: each features is measured fairly across all possible contexts.

Toward the SHAPley Value Formula

Let N enumerated as $\{1, 2, \ldots, n\}$ be a set of features.

How many possible orderings are there for N?

$|N|!$

Suppose S is a subset of features that arrived ***before*** feature i. State the number of orderings of S.

$|S|!$

S is sometimes called a ***coalition***.

Give an expression for the number of features that arrived ***after*** feature i. State the number of corresponding orderings.

$|N| - |S| - 1$ $\quad$ $(|N| - |S| - 1)!$

What does the expression $|S|! \cdot (|N| - |S| - 1)!$ represent?

The number of orderings where exactly the features in S arrive before feature i.

What does the expression $\frac{|S|! \cdot (|N| - |S| - 1)!}{|N|}$ represent?

The proportion of orderings where exactly the features in S arrive before feature i.

For a model f, express the marginal contribution of feature i: how much i contributes comparatively to the model's prediction when considering S.

$f(S \cup \{i\}) - f(S)$

The SHAPley Value Formula

If we combine the expressions to obtain the expression on the right, we observe that it computes

$$\frac{|S|! \cdot (|N| - |S| - 1)!}{|N|} \cdot [f(S \cup \{i\}) - f(S)]$$

the marginal contribution of i scaled by a fractional **weight**:
the number of sets with the number of orderings where exactly the features in S arrive before feature i.

If we sum the result of allowing feature i to sequentially arrive in each position in the ordering, we have the following definition.

Definition. Let N be a set of features enumerated by $\{1, 2, \dots, n\}$. Then, the ***Shapley value for feature i***, ϕ_i, is defined as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \cdot [f(S \cup \{i\}) - f(S)]$$

Weights

Consider a case where we have $|N| = 3$ features we will refer to as $N = \{C, I, D\}$.
This means $P(3,3) = 3! = 6$ orderings.

We will compute the weights for `Credit` score; Credit can arrive first, second or third.

Coalition S (before Credit)	$\lvert S\rvert$	Orderings	Count	Weight $\left(\frac{Count}{6}\right)$	$\frac{\lvert S\rvert!\,(\lvert N\rvert - \lvert S\rvert - 1)!}{\lvert N\rvert!}$
$\emptyset$	0	$C \to I \to D$ $C \to D \to I$	2	$\frac{1}{3}$	$\frac{0!\,(3-0-1)!}{3!} = \frac{1}{3}$
$\{I\}$	1	$I \to C \to D$	1	$\frac{1}{6}$	$\frac{1!\,(3-1-1)!}{3!} = \frac{1}{6}$
$\{D\}$	1	$D \to C \to I$	1	$\frac{1}{6}$	$\frac{1!\,(3-1-1)!}{3!} = \frac{1}{6}$
$\{I, D\}$	1	$I \to D \to C$ $D \to I \to C$	2	$\frac{1}{3}$	$\frac{2!\,(3-2-1)!}{3!} = \frac{1}{3}$

Computing ϕ_I

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \cdot [f(S \cup \{i\}) - f(S)]$$

For SHAP, we have to retrain a model for each coalition. We then can compute a prediction score for a single record in our data.

Suppose we then have the following prediction scores for a single applicant:
Income = \$85K, Credit = 720, Debt Ratio = 35%

For example, $f(\{C\}) = 0.65$ means: "For this applicant, using only their Credit Score (720), the model predicts 0.65 approval probability."

Coalition S	$f(S)$
$\emptyset$	0.50
$\{I\}$	0.58
$\{C\}$	0.65
$\{D\}$	0.42
$\{I, C\}$	0.72
$\{I, D\}$	0.52
$\{C, D\}$	0.60
$\{I, C, D\}$	0.78

Computing ϕ_I

Coalition S	$f(S)$	$S \cup \{I\}$	$f(S \cup \{I\})$
$\emptyset$	0.50	$\{I\}$	0.58
$\{C\}$	0.65	$\{I, C\}$	0.72
$\{D\}$	0.42	$\{I, D\}$	0.52
$\{C, D\}$	0.60	$\{I, C, D\}$	0.78

$$\phi_I = \frac{1}{3} \cdot (0.58 - 0.5) + \frac{1}{6} \cdot (0.72 - 0.65) + \frac{1}{6} \cdot (0.52 - 0.42) + \frac{1}{3} \cdot (0.78 - 0.60) = 0.115$$

Computing ϕ_C

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \cdot [f(S \cup \{i\}) - f(S)]$$

Compute $\phi_C = 0.19$ and $\phi_D = -0.025$.

Coalition S	$f(S)$
$\emptyset$	0.50
$\{I\}$	0.58
$\{C\}$	0.65
$\{D\}$	0.42
$\{I, C\}$	0.72
$\{I, D\}$	0.52
$\{C, D\}$	0.60
$\{I, C, D\}$	0.78

$$\phi_C = \frac{1}{3} \cdot (0.65 - 0.5) + \frac{1}{6} \cdot (0.72 - 0.58) + \frac{1}{6} \cdot (0.6 - 0.42) + \frac{1}{3} \cdot (0.78 - 0.52) = 0.19$$

$$\phi_D = \frac{1}{3} \cdot (0.42 - 0.5) + \frac{1}{6} \cdot (0.52 - 0.58) + \frac{1}{6} \cdot (0.6 - 0.65) + \frac{1}{3} \cdot (0.78 - 0.72) = -0.025$$

Altogether, ($\phi_C = +0.19$, $\phi_I = +0.115$) move this applicant from the 0.50 baseline to 0.78 approval probability. The poor Debt ratio did not help with approval ($\phi_D = -0.025$).

Observe in terms of overall explainability of this score for this applicant:

$$f(\{I, C, D\}) - f(\emptyset) = 0.78 - 0.50 = 0.28 = \phi_I + \phi_C + \phi_D$$

Properties of SHAP

Shapley values are the unique solution satisfying:

Axiom	Property
Efficiency	SHAP values sum to $f(\mathbf{x}) - base$. All credit distributed.
Symmetry	Equal contributors receive equal values.
Dummy	Non-contributing features receive zero.
Additivity	For $f + g$: $\phi_i(f + g) = \phi_i(f) + \phi_i(g)$

Shapley values are the *unique* attribution method satisfying all four properties. Any other method must violate at least one.*

*Shapley (1953) proved a uniqueness theorem.

The SHAP Process and Its Cost

To compute the SHAP value for ***one feature*** of $|N|$ features on ***one prediction***:

- Enumerate all coalitions S (subsets of other features)
- For each coalition, train or evaluate a model using only the features in S !
- Compute $f(S)$ and $f(S \cup \{i\})$ for each coalition
- Calculate the marginal contribution: $f(S \cup \{i\}) - f(S)$
- Weight each marginal by $\frac{|S|!(|N|-|S|-1)!}{|N|!}$
- Sum the weighted marginals: ϕ_i

For $|N|$ features, the set of all coalitions is the powerset: $2^{|N|}$ subsets. Each requires either training a separate model or simulating absent features by marginalizing over their values.

Exact SHAP is exponential in the number of features. For real-world models, we need approximations — KernelSHAP, TreeSHAP, DeepSHAP — that estimate Shapley values without exhaustive enumeration.

From Local to Global: SHAP Across a Dataset

Locally we ask: "Why was this applicant denied?"
Globally we ask: "What drives approvals in general?"

Definition. Let $\phi_i^{(j)}$ denote the SHAP value of feature i for sample j in a dataset of n samples. The ***global importance*** of feature i is defined as:

$$I_i = \frac{1}{n}\sum_{j=1}^{n}\left|\phi_i^{(j)}\right|$$

Interested in SHAP magnitude: features pushing predictions away from baselines *regardless of sign*.

Intuitively, I_i measures how much a single feature matters across all samples.

Low I_i → feature i rarely influences predictions.

High I_i → feature i consistently moves predictions away from the baseline, regardless of direction, across the dataset.

Visualizing Global SHAP: Beeswarm Plot

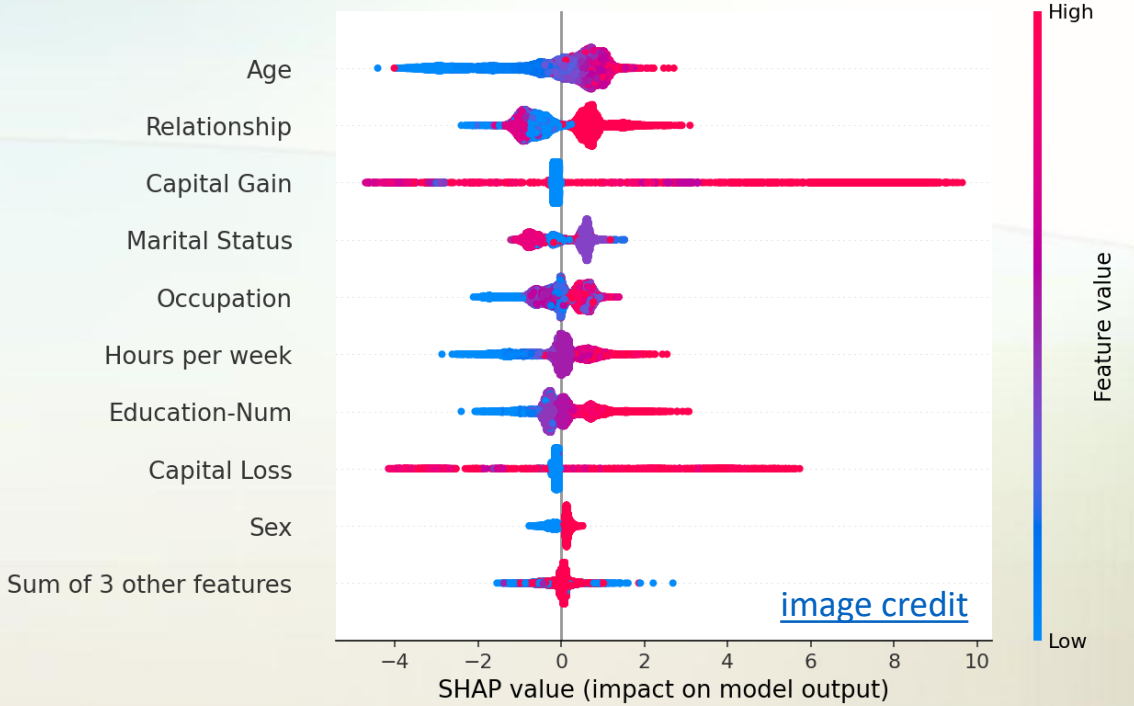
Adult income dataset (classification task to predict if people made over $50k in the 1990s).

X-axis: SHAP value (impact on prediction)

Y-axis: Features (ranked by importance: I_i)

Each dot represents one sample's SHAP value, $\phi_i^{(j)}$: how much feature i contributed for sample j.

Color indicates whether that sample had a low (blue) or high (red) value for the feature.



Age

Red dots ≡ older individuals
Blue dots ≡ younger individuals

Age affects many samples moderately (spread of dots)

Capitol Gain affects few samples, but those few see a large effect (tight cluster + long tail)

Most people have little or no capital gain, so the feature does not affect their prediction. But for those who do have high capital gains, the impact is dramatic.